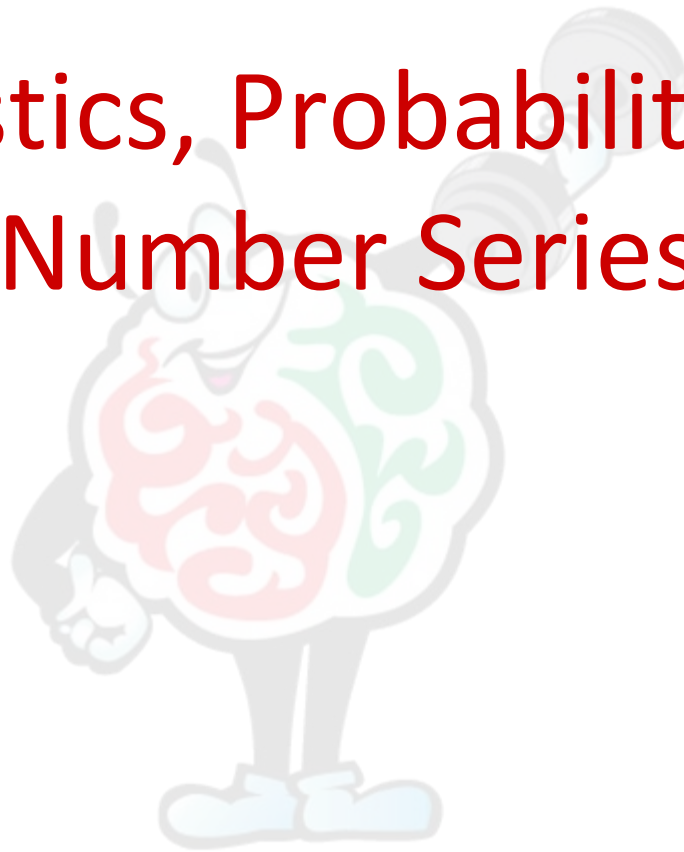


Statistics, Probability and Number Series



Measures of central tendency

- Mean

- Add the numbers and divide the sum by the number of values (which were added).

- Median

- The middle number when numbers are written in order. (When the number of values is even, the median is the mean of the two middle values.)

- Mode

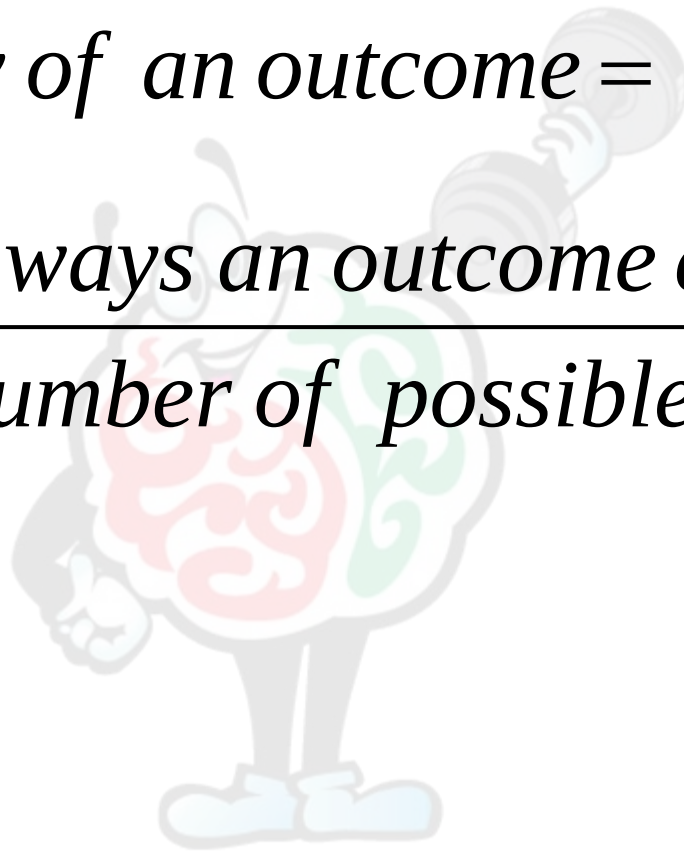
- The number which occurs most often

PRACTICE

- May took 7 maths tests and got scores of 68, 71, 71, 84, 53, 62 and 67. What was May's mode score?
- The mean of three numbers is 8. Two of the numbers are 11 and 7. What is the third number?
- The cost of five cakes is P28, P19, P45, P45, P15. What is the median cost?

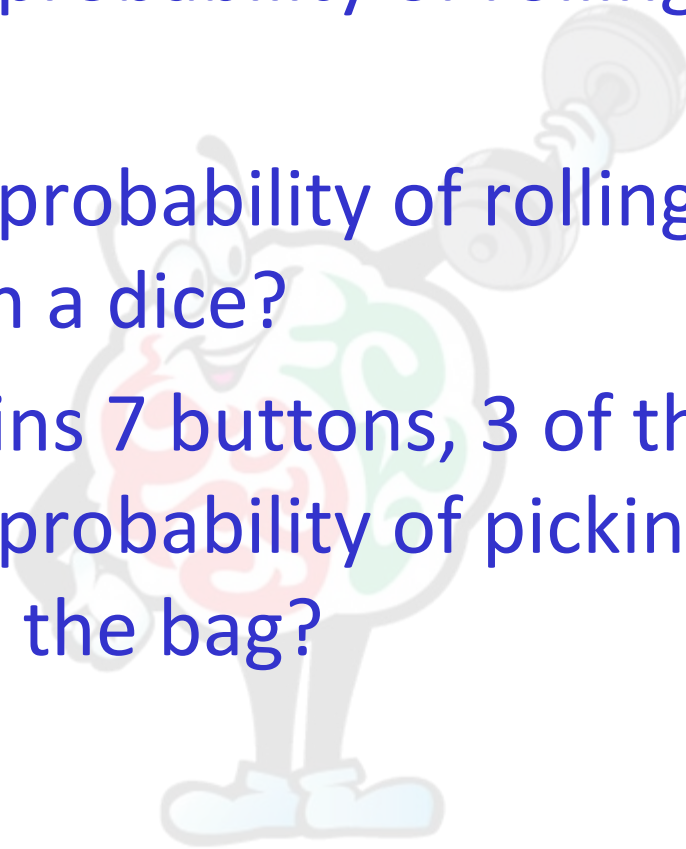
Probability

Probability of an outcome =
number of ways an outcome can happen
the total number of possible outcomes



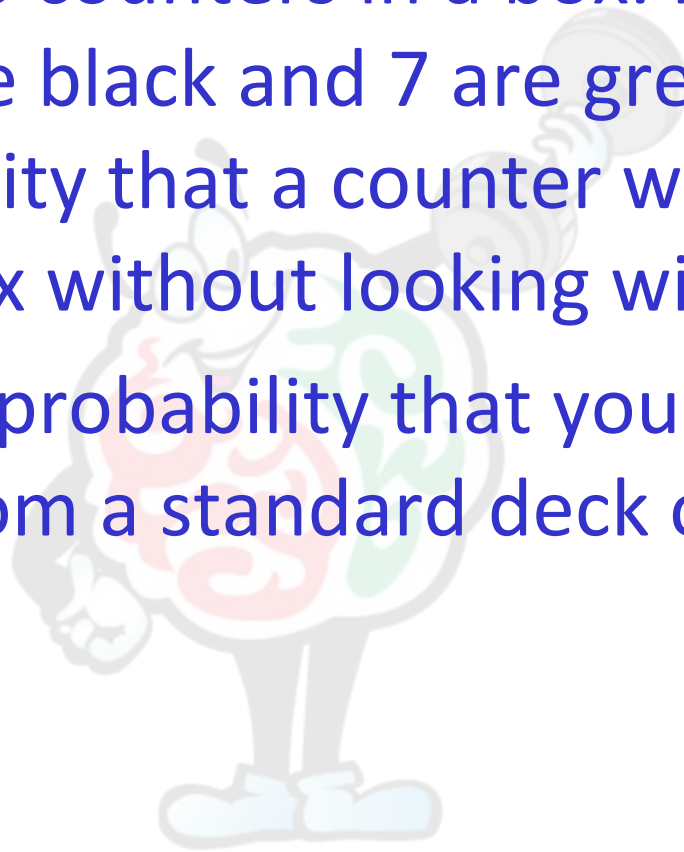
PRACTICE

- What is the probability of rolling a 4 with a dice?
- What is the probability of rolling an odd number with a dice?
- A bag contains 7 buttons, 3 of them green. What is the probability of picking a green button from the bag?



PRACTICE

- There are 20 counters in a box. Nine are yellow, 4 are black and 7 are green. What's the probability that a counter which is taken from the box without looking will be green?
- What is the probability that you will draw a face card from a standard deck of playing cards?



Counting techniques

- If two or more actions are performed in a particular order, such that the first action produces a results, the second action produces b results, the third action produces c results, and so on, then:

$a * b * c * \dots = \text{total number of possible results}$

Counting techniques: Permutation

- Objects are arranged in a definite order:

- n different things taken r at a time

$${}_n P_r = \frac{n!}{(n - r)!}$$

- n things taken n at a time in which q are alike, r are alike and so on

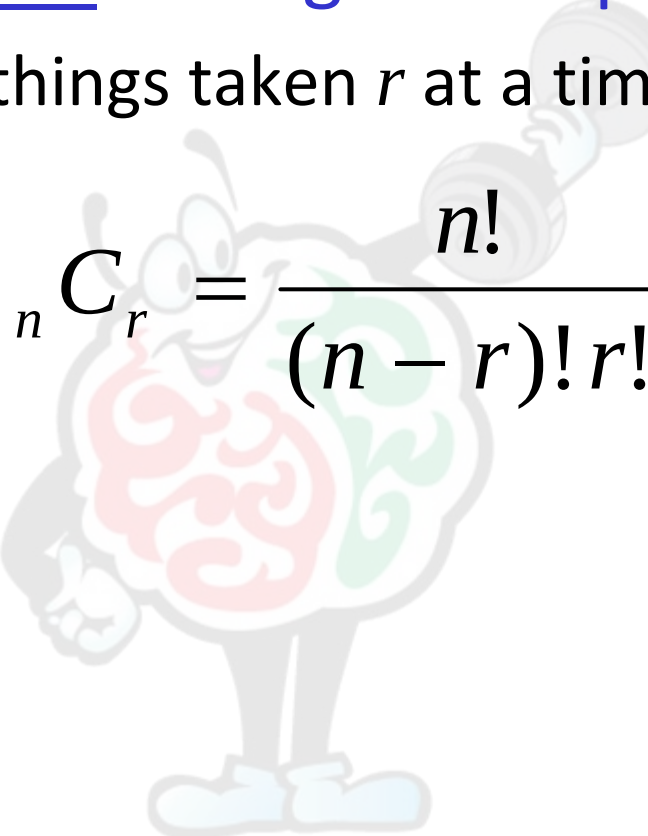
$${}_n P_r = \frac{n!}{(q! r! \dots)}$$

- n distinct objects are arranged along a fixed circle

$$P_n = (n - 1)!$$

Counting techniques: Combination

- Objects are not arranged in a particular order
 - n different things taken r at a time

$${}_nC_r = \frac{n!}{(n-r)!r!}$$


Arithmetic sequences and series

- Arithmetic sequence/progression: a sequence of numbers where successive members of the sequence have a constant difference (d).
- Formula for n -th term $a_n = a_1 + (n - 1)d$
- Formula for sum of first n integers (or sum of integers from 1 to n)

$$S_n = \frac{n(n + 1)}{2}$$

Geometric sequences and series

- Geometric sequence/progression: a sequence of numbers where successive members of the sequence are produced by multiplying them by a constant multiplier (r , or common ratio)
- Formula for n -th term $a_n = a_1 * r^{n-1}$
- Formula for sum of first n values of a sequence

$$S_n = \frac{a_1(1 - r^n)}{1 - r}$$

References and further reading

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